

Arizona State University
School of Electrical, Computer and Energy Engineering

EEE 465/591: Photovoltaics Energy Conversion
Practice Problem Set #4

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In this practice problem set, we focus on carrier lifetimes as well as calculations and drawing the band diagram of pn junctions.

1.

Given the material parameters below, calculate the effective lifetime and say which recombination mechanism dominates.

$$\tau_{\text{rad}} = 10 \mu\text{s}, \tau_{\text{SRH}} = 20 \mu\text{s}, \tau_{\text{Auger}} = 0.5 \text{ ms}$$

2.

For the following material parameters, **estimate**: (1) which version of the saturation current you should use (e.g., wide base, narrow base, or generalized); and (2) if you would except the electron or hole diffusion current to be larger (or if they are the same). The pn junction is made from silicon, with $n_i = 1 \times 10^{10} \text{ cm}^{-3}$ and E_G (at 300K) = 1.12 eV. **You don't need to calculate each case, look at the magnitudes of the parameters.**

N-type material parameters

$$N_D = 1 \times 10^{16} \text{ cm}^{-3}; D_p = 15 \text{ cm}^2/\text{s}; \text{Thickness} = 20 \mu\text{m}; \tau_p = 1000 \mu\text{s}; S = 1 \times 10^6 \text{ cm/sec}$$

P-type material parameters

$N_A = 5 \times 10^{18} \text{ cm}^{-3}$; $D_n = 40 \text{ cm}^2/\text{s}$; Thickness = $200 \text{ }\mu\text{m}$; $\tau_n = 100 \text{ }\mu\text{s}$; $S = 10 \text{ cm/sec}$

3.

Calculate the saturation current density and the current density at a bias of 0.65V.

4.

Sketch the band diagram at (1) equilibrium and (2) a forward bias of 0.65V. You should sketch the quasi-Fermi level.

1. Effective Lifetime & Dominant Recombination

Combine recombination channels like parallel rates:

$$1/\tau_{\text{eff}} = 1/\tau_{\text{rad}} + 1/\tau_{\text{SRH}} + 1/\tau_{\text{Auger}}$$

Calculation:

$$\begin{aligned} 1/\tau_{\text{eff}} &= 1/(10 \text{ }\mu\text{s}) + 1/(20 \text{ }\mu\text{s}) + 1/(0.5 \text{ ms}) \\ &= 1/(10 \text{ }\mu\text{s}) + 1/(20 \text{ }\mu\text{s}) + 1/(500 \text{ }\mu\text{s}) \\ &= 0.1 + 0.05 + 0.002 = 0.152 \text{ (}\mu\text{s)}^{-1} \end{aligned}$$

$$\tau_{\text{eff}} = 1/0.152 \approx 6.58 \text{ }\mu\text{s}$$

Dominant Mechanism: The shortest lifetime dominates the rate.

Since $\tau_{\text{rad}} = 10 \text{ }\mu\text{s} < \tau_{\text{SRH}} = 20 \text{ }\mu\text{s} < \tau_{\text{Auger}} = 500 \text{ }\mu\text{s}$

Radiative recombination dominates

2. Which I_s Form and Which Diffusion Current is Larger?

Step 1: Calculate Diffusion Lengths

Minority holes in N region:

$$L_p = \sqrt{D_p \tau_p} = \sqrt{(15 \times 10^{-3})} = 0.122 \text{ cm} = 1220 \text{ } \mu\text{m}$$

Minority electrons in P region:

$$L_n = \sqrt{D_n \tau_n} = \sqrt{(40 \times 10^{-4})} = 0.063 \text{ cm} = 632 \text{ } \mu\text{m}$$

Step 2: Compare W and L

N side: $W_n/L_p = 20/1220 \approx 0.016 \ll 1 \Rightarrow$ narrow-base

P side: $W_p/L_n = 200/632 \approx 0.32 < 1 \Rightarrow$ narrow-base

Step 3: Include Surface Recombination Effects

N side (holes): $D_p/S = 15/10^6 = 1.5 \times 10^{-5} \text{ cm} = 0.15 \text{ } \mu\text{m}$

$W_n + D_p/S \approx 20.15 \text{ } \mu\text{m} \Rightarrow$ strong surface recombination

P side (electrons): $D_n/S = 40/10 = 4 \text{ cm} = 40,000 \text{ } \mu\text{m}$

$W_p + D_n/S \approx 4.02 \text{ cm} \gg W_p \Rightarrow$ very weak surface recombination

Step 4: Which Diffusion Current is Larger?

$$J_p \propto D_p/[N_D(W_n + D_p/S)] \approx 15/[(1 \times 10^{16})(0.002015)] \approx 7.44 \times 10^{-13}$$

$$J_n \propto D_n/[N_A(W_p + D_n/S)] \approx 40/[(5 \times 10^{18})(4.02)] \approx 2.0 \times 10^{-18}$$

Hole diffusion current from N side dominates by $\sim 3.7 \times 10^5$

3. Saturation Current Density and Current at 0.65V

$$J_s = qn_i^2 \left[\frac{D_p}{N_D(W_n + D_p/S_n)} + \frac{D_n}{N_A(W_p + D_n/S_p)} \right]$$

Given (300K):

$$n_i = 10^{10} \text{ cm}^{-3}, q = 1.602 \times 10^{-19} \text{ C}$$

$$N_D = 1 \times 10^{16}, D_p = 15 \text{ cm}^2/\text{s}, W_n = 0.002 \text{ cm}, S_n = 10^6 \text{ cm/s}$$

$$N_A = 5 \times 10^{18}, D_n = 40 \text{ cm}^2/\text{s}, W_p = 0.02 \text{ cm}, S_p = 10 \text{ cm/s}$$

Effective thickness adjustments:

$$D_p/S_n = 1.5 \times 10^{-5} \text{ cm} \rightarrow W_n + D_p/S_n = 0.002015 \text{ cm}$$

$$D_n/S_p = 4 \text{ cm} \rightarrow W_p + D_n/S_p = 4.02 \text{ cm}$$

Calculate J_s :

$$D_p/[N_D(W_n + D_p/S_n)] = 15/[(10^{16})(0.002015)] = 7.44 \times 10^{-13}$$

$$D_n/[N_A(W_p + D_n/S_p)] = 40/[(5 \times 10^{18})(4.02)] = 2.0 \times 10^{-18}$$

$$J_s \approx q n_i^2 (7.44 \times 10^{-13}) \approx 1.19 \times 10^{-11} \text{ A/cm}^2$$

Current density at $V = 0.65\text{V}$:

$$V_T = kT/q \approx 0.02585 \text{ V}$$

$$J(V) = J_s(e^{V/V_T} - 1)$$

$$e^{0.65/0.02585} \approx 8.3 \times 10^{10}$$

$$J(0.65\text{V}) \approx 0.99 \text{ A/cm}^2$$

4. Band Diagrams

Built-in Potential:

$$V_{bi} = V_T \ln(N_A N_D / n_i^2)$$

$$= 0.02585 \ln[(5 \times 10^{18})(10^{16})/10^{20}]$$

$$V_{bi} \approx 0.875 \text{ V}$$

(a) Equilibrium (no bias):

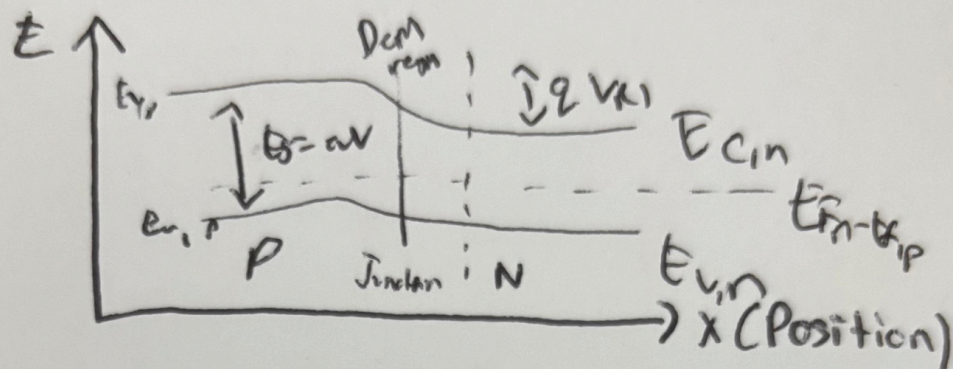
- E_C and E_V bend by $qV_{bi} \approx 0.875$ eV across the depletion region
- Single flat Fermi level E_F everywhere
- Depletion partitions mainly into lightly-doped n side ($W_n/W_p \approx 22.4$)
- Most band bending occurs on the n side

(b) Forward bias $V_A = 0.65$ V:

- Barrier lowers to $V_{bi} - V_A \approx 0.225$ V
- Smaller band step across depletion region
- **Quasi-Fermi levels split:**
 - E_{Fn} : nearly flat across neutral n region
 - E_{Fp} : nearly flat across neutral p region
 - $E_{Fn} - E_{Fp} \approx qV_A = 0.65$ eV across junction
- Minority carrier edges rise by e^{V_A/V_T}
- Stronger hole injection from N side (dominant contribution)
- Much weaker electron injection from P side (due to large N_A and small S_p)

4.)

a) Equilibrium



b) Forward Bias $V = .65\text{ V}$

